QISKIT FALL FEST 2026 · IIIT-DELHI · TRACK 3 QUANTUM SIMULATION · INTERMEDIATE + ADVANCED ANALYSIS

Dynamics Under Noise

Trotter errors with 8 spins, error mitigation with 12, and a 100-qubit kicked Ising chain checked against exact answers

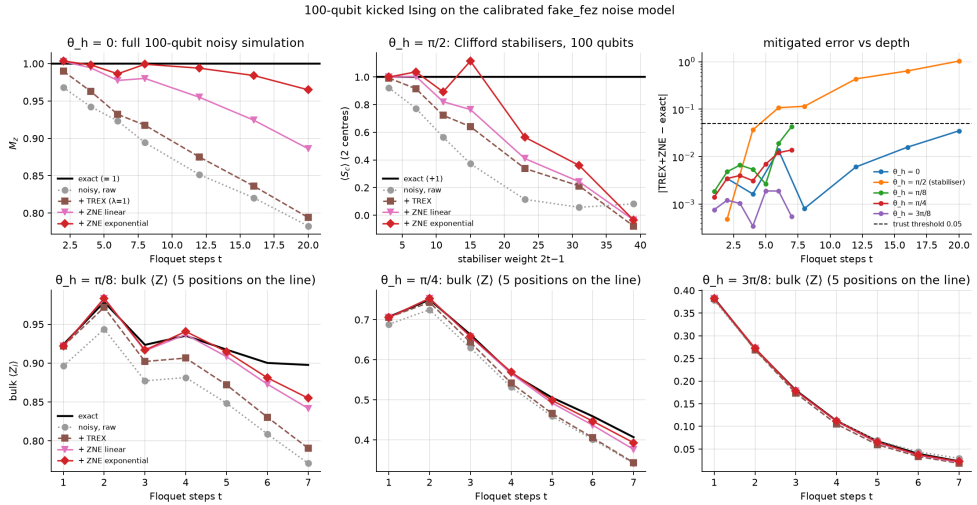
100 qubits
error-aware line on heavy-hex,
0 SWAPs

1881 CZ
deepest circuit, 2q depth 38, 20 Floquet steps

3 exact checks
free fermions + Pfaffians · light cone · Clifford, all at N = 100

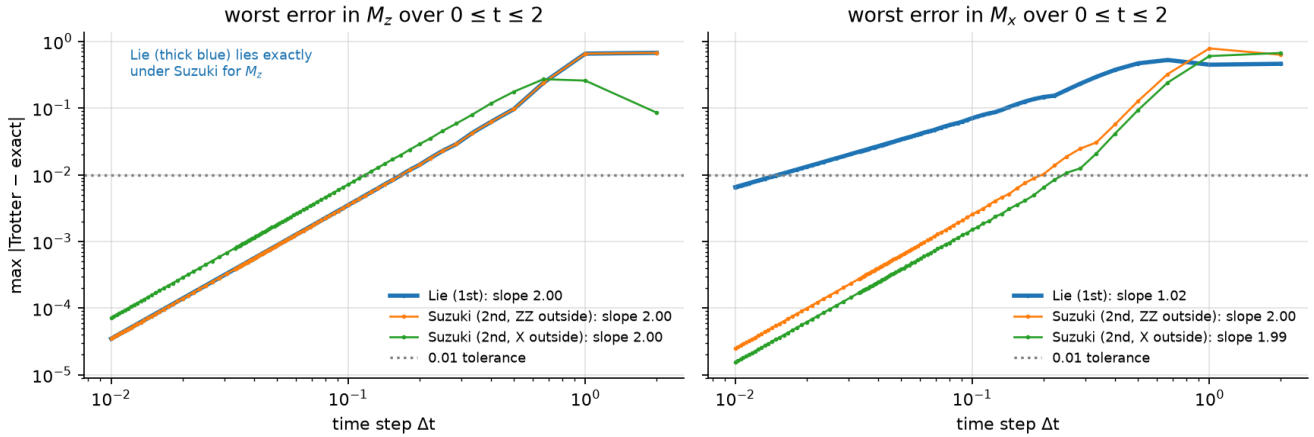
t = 20 recovered
θ_h=0 M_z within 0.05 of exact at 1881 CZ (100-qubit noisy emulation); stabilisers only to t = 4 (weight ≤ 7)

- **Beginner:** for Z observables from |0…0⟩, Lie-Trotter and Suzuki-Trotter give identical results. For Mx, 2nd order needs 12× fewer CX for 0.01 accuracy (154 vs 1834).
- **Intermediate:** 2nd-order accuracy for the CZ cost of 1st order, our own TREX + ZNE on FakeFez noise, and the finding that under noise finer Trotter steps stop helping.
- **Advanced:** 100-qubit kicked Ising (0 SWAPs) on a noise model from the chosen line's calibration, with TREX + ZNE, compared with exact answers at every depth. The QPU jobs are prepared and validated, but we had no QPU access before the deadline.



BEGINNER · 8 SPINS · J = h = 1 · t ≤ 2

Why the "right" Trotter step depends on what you measure



formula	obs	largest Δt (err<0.01)	CX Qiskit	CX merged
lie	Mz	0.1667 (12 steps)	168	154
lie	Mx	0.0152 (132 steps)	1848	1834
suzuki	Mz	0.1667 (12 steps)	336	168
suzuki	Mx	0.1818 (11 steps)	308	154
strang	Mz	0.1176 (17 steps)	442	238
strang	Mx	0.2222 (9 steps)	234	126

- $|0\ldots0\rangle$ is an eigenstate of every Z_iZ_{i+1}, so the **first ZZ layer is a phase**.
- A **final ZZ layer commutes** with every Z-basis observable.
- Without those two pieces, Lie and Suzuki apply the same gates for M_z and $\langle Z_0Z_r\rangle$ (difference 1e-16), so Lie is effectively 2nd order (slope 2.00).
- For M_x the final ZZ(Δt/2) is the correction: Lie is 1st order (slope 1.02), Δt = 0.015 vs 0.18.
- Merging neighbouring half-steps gives 2nd order for the CX cost of 1st order.

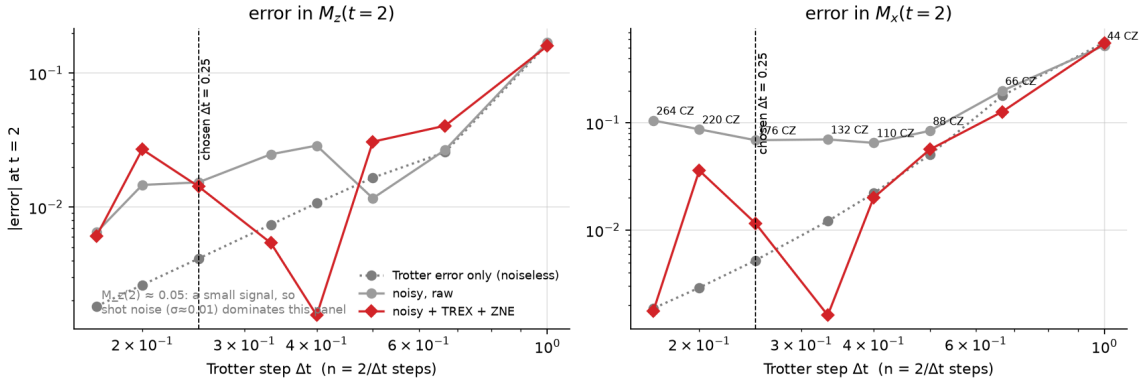
> Cheapest correct choice: Lie for Z-only; merged Suzuki (Δt = 2/11, 154 CX) for M_x; Δt = 2/12 (168 CX) for both.

INTERMEDIATE · SHORTER CIRCUITS

Shorter circuits first: fewer CZs means less error to mitigate

12-spin TFIM to t = 2 on FakeFez	CZ	2q depth	SWAPs	ESP
(a) Qiskit SuzukiTrotter, default layout	322	322	0	0.298
(b) merged brickwork, default layout	176	32	0	0.499
(c) merged brickwork, error-aware line	176	32	0	0.499
(d) merged brickwork, scattered qubits	747	184	190	0.000

- Brickwork ordering (even bonds, then odd), so the 2q depth per step does not grow with N.
- Merged half-steps, and the trivial first layer removed.
- Error-aware line: a beam search over the coupling map that minimises the summed CZ, readout and SX infidelity from calibration data. It finds a 100-qubit path on the 156-qubit Heron.
- Kicked Ising at $\theta_J = -\pi/2$: each RZZ is a single CZ. 100 qubits, 20 steps: 1881 CZ, 0 SWAPs.



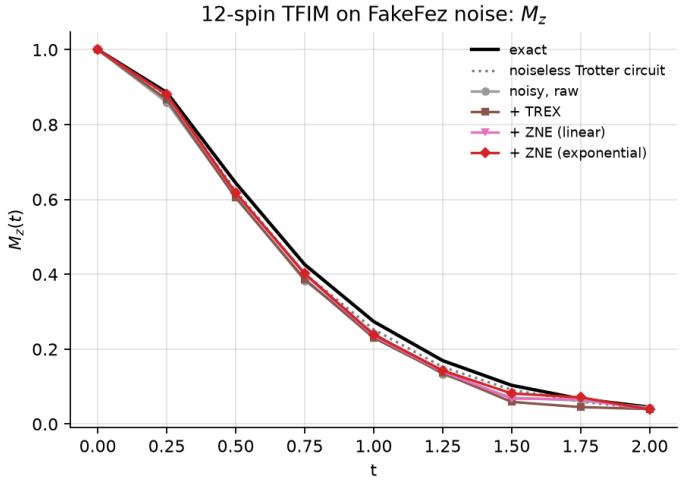
For Δt ≈ 0.25–0.4 the raw error levels off at about 0.07, and finer steps make it worse (0.105 at Δt = 0.167). After ZNE all Δt ≤ 0.4 agree within shot noise, so the Trotter-only choice Δt ≈ 0.15 gains nothing and costs 62% more CZ (286 vs 176).

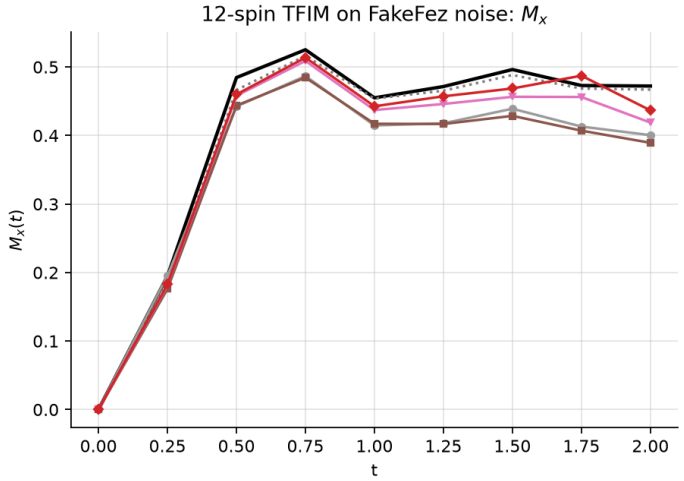
ESP = Π(1−ε_gate)·Π(1−ε_readout) from FakeFez calibration

§2.1, §2.4

INTERMEDIATE · 12 SPINS · FakeFez NOISE · OUR OWN MITIGATION CODE

Error mitigation on the 12-spin chain





- TREX: random X before readout, undone in post-processing, then rescaled using a twirled $|0\ldots0\rangle$ calibration.
- ZNE: every CZ folded into 3 or 5 CZs, then a linear, exponential or Richardson fit to $\lambda \to 0$.
- Twirling makes no difference on Pauli noise (we checked). It matters on hardware, where some errors are coherent.
- Compact noise model checked against the full FakeFez model.

mean |error| vs exact	M_z	M_x
raw	0.027	0.045
TREX	0.031	0.051
TREX + ZNE (linear)	0.021	0.026
TREX + ZNE (exp)	0.019	0.019

Job B (prepared, not run) would re-run these 8 circuits on a QPU for the noisy-sim vs hardware comparison

ADVANCED · HOW DO WE KNOW THE ANSWER AT 100 QUBITS?

Three exact methods that work at 100 qubits

method	what it gives	cost	valid
Free fermions + Pfaffians JW in X basis, a ghost spin turns Z_i into an even Majorana string; Wick → Pfaffian	every $\langle Z_i \rangle$, any $\langle Z_i Z_j \rangle$	$O(N^3)$ · 0.3 s at N = 100, t = 20	integrable 1D kicked TFIM, **any depth**
Light-cone window $\langle Z_k(t) \rangle$ depends only on [k−t, k+t]; one (2t+2)-qubit edge window + reflection gives all 100 sites	every $\langle Z_i \rangle$, $\langle Z_i Z_{i+1} \rangle$	2^{2t+2}, independent of N	any local circuit, $t \lesssim 11$
Clifford point $\theta_h = \pi/2$ $S = U Z_c U^\dagger$, weight 2t−1 Pauli	$\langle S \rangle = +1$ exactly	stabiliser tableau	any N, any depth

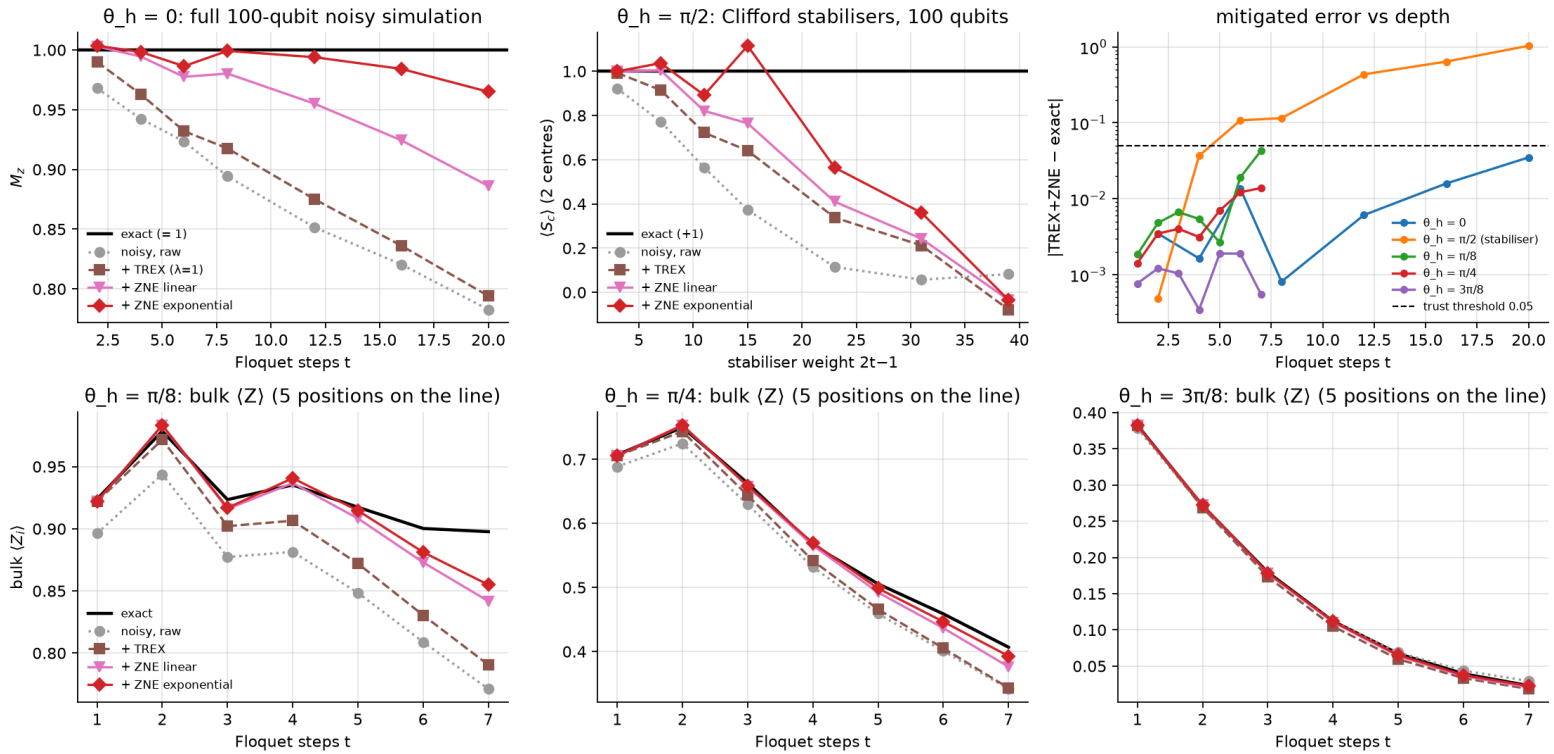
- Checked against each other: free-fermion and light-cone results at N = 100 agree to about 1e-14, and both match statevector at N = 12.
- Kicked Ising (Kim *et al.*, Nature 2023) on a 1D chain: $U_F = \Pi\, R_X(\theta_h) \cdot \Pi\, R_{ZZ}(-\pi/2)$.
- 52 circuits: $\theta_h \in \{0, \pi/8, \pi/4, 3\pi/8\} \times t \in \{1 \ldots 8, 10, 12, 14, 16, 20\}$, plus 7 Clifford circuits.

> Caveat: integrability is what lets us check 100 qubits at depth 20, and it also makes this model easy for a classical computer. We use it as a test bench with known answers, not as a quantum-advantage claim.

ADVANCED · 100 QUBITS · CALIBRATED IBM HERON NOISE MODEL

100 qubits on a noise model: raw, mitigated and exact

100-qubit kicked Ising on the calibrated fake_fez noise model

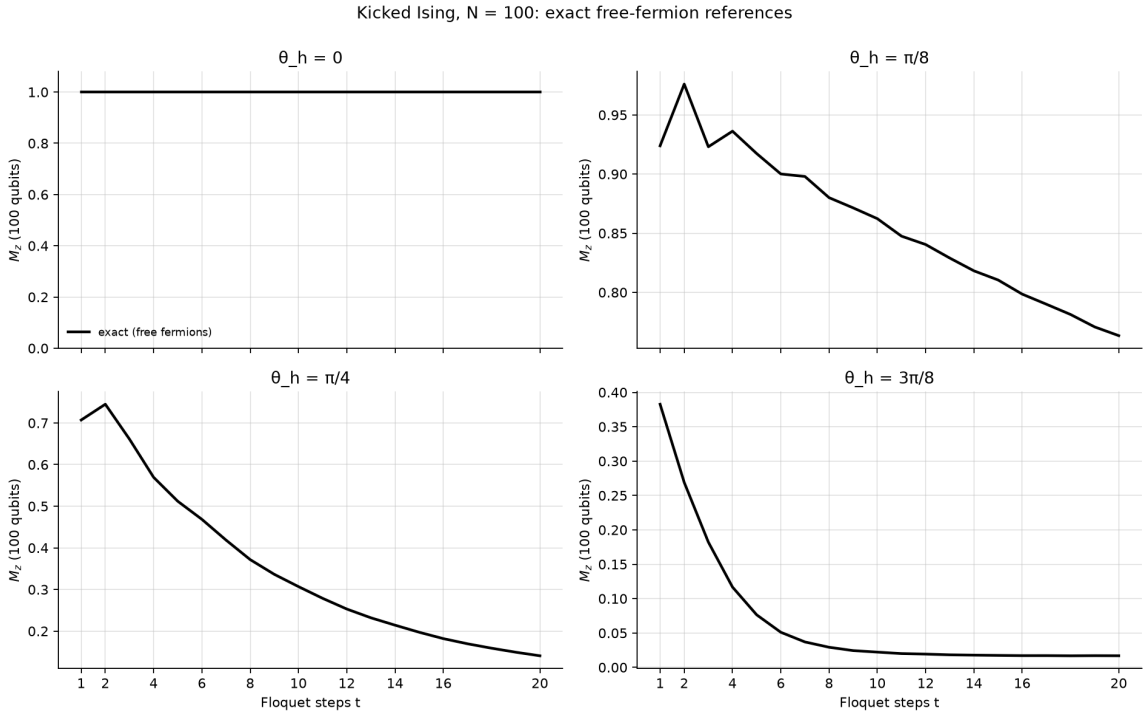


- **Clifford points** ($\theta_h = 0, \pi/2$): a stabiliser simulation of the whole noisy 100-qubit circuit is exact, so we run it at full size up to 20 steps (1881 CZ), including the folded ZNE circuits.
- **Other angles**: noise outside the light cone cannot affect the result, so we simulate (2t+1)-qubit windows at 5 positions on the line. They agree with full-chain density matrices to $\leq 4\times10^{-4}$.
- Noise: per-gate CZ/SX errors + asymmetric readout of the *chosen* 100 qubits (ibm_fez snapshot).

θ_h=0, t=20 (1881 CZ): raw 0.78 → TREX+ZNE 0.96 (exact 1); weight-39 stabiliser: raw 0.08 → -0.03 (exact 1).

ADVANCED · TRUST HORIZON · QPU-READY

How far can we trust it, and how would we know without an exact answer?



- **Trust horizon** (|ZNE − exact| ≤ 0.05): 0: t=20, π/2 (stabiliser): t=4, π/8: t=7, π/4: t=7, 3π/8: t=7.
- **Which agreements count**: noise pulls ⟨Z⟩ towards 0, so $\theta_h = 3\pi/8$ (exact ≈ 0) "agrees" anyway. The observables with a large signal ($\theta_h = 0$, π/8, Clifford stabilisers) are the real test.
- **Without an exact answer**: exponential vs linear ZNE spread, Clifford-point decay and light-cone windows (t ≲ 11).
- **QPU-ready**: 3 jobs (EstimatorV2 with TREX + twirling + XpXm DD + ZNE; SamplerV2 raw + readout calibration), PUBs validated, ≈ 4 Open Plan minutes (estimate); a ~1–2 min `--quick` subset. Every CZ count equals (t−1)·99, proving 0 SWAPs.

Main lesson: how far we can trust a result depends on the observable's weight. A single ⟨Z⟩ after 1881 CZ comes back to 0.96 (exact 1), but a Pauli spread over 15+ qubits sinks below the shot noise at λ = 5 and ZNE cannot recover it. Weight-7 stabilisers pass; weight 11 and above fail.

exact $M_z(t)$ for the 4 angles on the left; free-fermion Pfaffians, 0.3 s each

LIMITATIONS · NEXT STEPS · REPRODUCIBILITY

Limitations and next steps

- **Classically easy model**: integrability is what lets us check the results exactly. A longitudinal field h_z would break it, leaving only the light-cone, Clifford and consistency checks.
- **Gate-folding ZNE** makes circuits 3–5× deeper. At $t \geq 16$ (1500+ CZ) the $\lambda = 5$ data is near the noise floor and the fits start to disagree.
- **No real-QPU data**: the noise model has no coherent errors, crosstalk or drift, so real hardware would decay faster. Twirling and DD are aimed at exactly those effects.
- **Next**: PEA-based ZNE with learned layer noise (our circuits have only 2 distinct CZ layers), fractional RZZ for the TFIM (half the depth), and a non-integrable chain.

prepared QPU job	primitive	content
A	EstimatorV2: TREX, twirl, DD, ZNE	52 kicked-Ising + 7 Clifford circuits, 100 qubits
B	EstimatorV2: same options	8 TFIM circuits, 12 qubits
C	SamplerV2 (raw)	26 circuits + readout calibration

- One notebook that runs top to bottom in about 5 min; the long sweeps are cached in `results/`.
- `hardware/run_hardware.py`: dry-run / submit / fetch with the exact same circuits.
- Tutorial credit: BasQ Qiskit Fall Fest 2026 (B. Tirado). AI tools: Claude was used for coding and writing help; all numbers are checked in the notebook.